

Application of NPP-VIIRS Nighttime Light Imagery for GDP Spatialization and Economic Differences in Vietnam

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This research aims to investigate whether nighttime light imagery can be used to estimate and spatialize regional GDP in Vietnam. By adapting the method proposed by Zhao et al. (2017), this research will use NPP-VIIRS nighttime light data and regional GDP statistics to evaluate its ability to capture spatial variations in economic activity. In addition, this research will develop an automated and reproducible analytical workflow that takes NPP-VIIRS imagery and regional GDP data as inputs and produces spatial GDP estimates. The study will then apply the workflow to Vietnam's GDP data to assess its effectiveness in capturing regional economic differences. The expected outputs include a spatial GDP dataset and map for Vietnam, an automated tool for GDP estimation, and a research paper documenting the methodology and findings, providing additional support for future regional socioeconomic research.

In order to successfully complete this project, I will apply my existing coding and data analysis skills, including collecting data from open sources, performing data cleaning, and developing the analytical tool to create a spatial GDP map and data for Vietnam. One of the main challenges I expect is learning how to process nighttime light imagery, which will likely be the most complex part of the project. To address this challenge, I will use online resources and tutorials on image processing and ask for guidance from faculty in the Economics Department to verify the accuracy and credibility of the methods used.

Gross domestic product (GDP) serves as a fundamental indicator of economic performance within a country or region. However, conventional socioeconomic statistics used to measure

GDP often suffer from inconsistent statistical scales and limited spatial resolution, making it difficult to capture regional economic disparities at a fine spatial scale. Elvidge et al. (1997) found a strong correlation between the area lit and GDP across 21 counties in North America using DMSP-OLS nighttime light data. Similar studies using data from the United States, Europe, and China have also reported strong relationships between total nighttime light and GDP (Doll et al., 2006; Zhao et al., 2011; Forbes, 2013). For data sources, this research will utilize freely available NPP-VIIRS nighttime light imagery. NPP-VIIRS provides higher spatial resolution (approximately 500 m) than DMSP-OLS data (1 km), while its wider radiometric detection range reduces saturation and its onboard radiometric calibration improves data quality (Elvidge et al., 2013).

Starting next week, I will begin the literature review and finalize the research methodology. By September 27, I aim to complete the procedure for estimating GDP using NPP-VIIRS data and begin applying it to real-world data. I plan to obtain preliminary results by October 18, allowing time to evaluate and refine the analysis with feedback from professors and colleagues. Based on their advice, I will conduct further analysis and improve the results. I aim to complete the first draft of the final paper by November 8, then revise it through peer review. The final paper and poster will be completed after incorporating peer-review feedback. A detailed description of each week's work is provided below.

Assignment	Official deadline
Literature Review	Week 4 (by September 20)
Research Design	Week 5 (by September 27)
Revised Design	Week 6 (by October 4)
Early Results & Presentations	Week 7-9 (by October 25)

First Draft	Week 10-11 (by November 8)
Peer Review	Week 12 (by November 15)
Executive Summary	Week 13-15 (by December 6, include Thanksgiving break)
Final Paper	Week 16 (by December 13)
Poster	Week 17 (December 12–17)

References

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